$H \to WW$ + charm

Analysis status

Chirayu Gupta · VUB · August 2026

2022postEE · 26.7 fb^{-1} · expected UL 1034

Sections

§	topic
1	Analysis and selection
2	MVA-defined regions
3	c-tagging — 2D working points and scale factors
4	Negative-weight reweighting
5	W+jets jet-binned samples
6	Systematics and impacts
7	Status and outlook

Dedicated decks: `ctag-sf-deck.pdf` (34 slides) · `negrw-deck.pdf` (57 slides)

1 · Analysis and selection

Why H+c

The Higgs coupling to **charm** is the least constrained of the accessible Yukawas. Associated production **H + c** probes it directly, without relying on the $H \to c\bar{c}$ decay.

production	pp → H + c (+X)
decay	H → WW* → 2ℓ2ν
final state	**opposite-sign eμ** + MET + ≥1 c-tagged jet
dataset	2022postEE, **26.7 fb^{-1}**, ReReco `22Sep2023`

> The eμ requirement removes Z→ee/μμ by construction, so the irreducible background is **real $t\bar{t}$** rather than Drell-Yan.

Reference: **AN-23-102**, the full Run 2 analysis at 138 fb^{-1}.

Signal and final state

$H \to WW \to 2\ell 2\nu$, produced with a charm quark.

Final state: **opposite-sign eμ** + MET + **≥1 c-tagged jet**

object	selection
muons	tight ID + tight PF iso, $p_T > 10$, $\lvert\eta\rvert < 2.4$
electrons	wp80iso, $p_T > 10$, $\lvert\eta\rvert < 2.5$, $\Delta R(e,\mu) > 0.4$
ll pair	OS, $p_T > 20 / 10$, pair $p_T > 30$, $m_{ll} > 12$
jets	$p_T > 30$, $\lvert\eta\rvert < 2.4$, tight-lepveto ID, $\Delta R(j,\ell) > 0.4$
c-jets	$p_T > 20$, **PNet medium** CvL/CvB
MET	PuppiMET

eμ removes the $Z \to ee/\mu\mu$ peak **by construction** — the irreducible background is **real $t\bar{t}$**, not Drell-Yan. Triggers: MuonEG, SingleMu, SingleEle.

Signal region composition

process	SR yield	share of SR
$t\bar{t}$	17,744	82.3%
st	1,543	7.2%
vjets	1,523	7.1%
diboson	609	2.8%
higgsbkg	132	0.6%
H+c signal	**0.26**	**0.001%**
total	**21,552**	

The signal is **0.26 events** against ~21,600 background — which is why MC statistics, rather than data statistics, drives the sensitivity.

2 · MVA-defined regions

Six classes, one network

setting	value
model	`SimpleMLP_MultiClass` (b-hive)
classes	`hplusc, higgsbkg, tt, st, diboson, vjets`
features	**26**, including the **11 c-tag one-hot categories**
training	30 epochs, batch 1024, lr 1e-3, loss weighting on
split	80/20, deterministic by NanoAOD `event` id

> The 11 one-hot categories mean the network sees the **full 2D CvL/CvB plane** —
> the same information the scale factors are binned in, not a binary pass/fail.

argmax defines the regions

The winning class picks the channel; the winning score is the discriminant.

region	events	dominant process
SR_hplusc	21,552	**tt 82.3%**
CR_higgsbkg	9,220	tt 86.4%
CR_tt	44,199	**tt 94.1%**
CR_st	20,135	tt 87.6%
CR_diboson	6,585	tt 72.4%
CR_vjets	9,214	**vjets 43.2%**

Regions are **orthogonal by construction** — no cut optimisation to tune or defend, and every event is used somewhere.

Four of five CRs are tt-dominated — by design

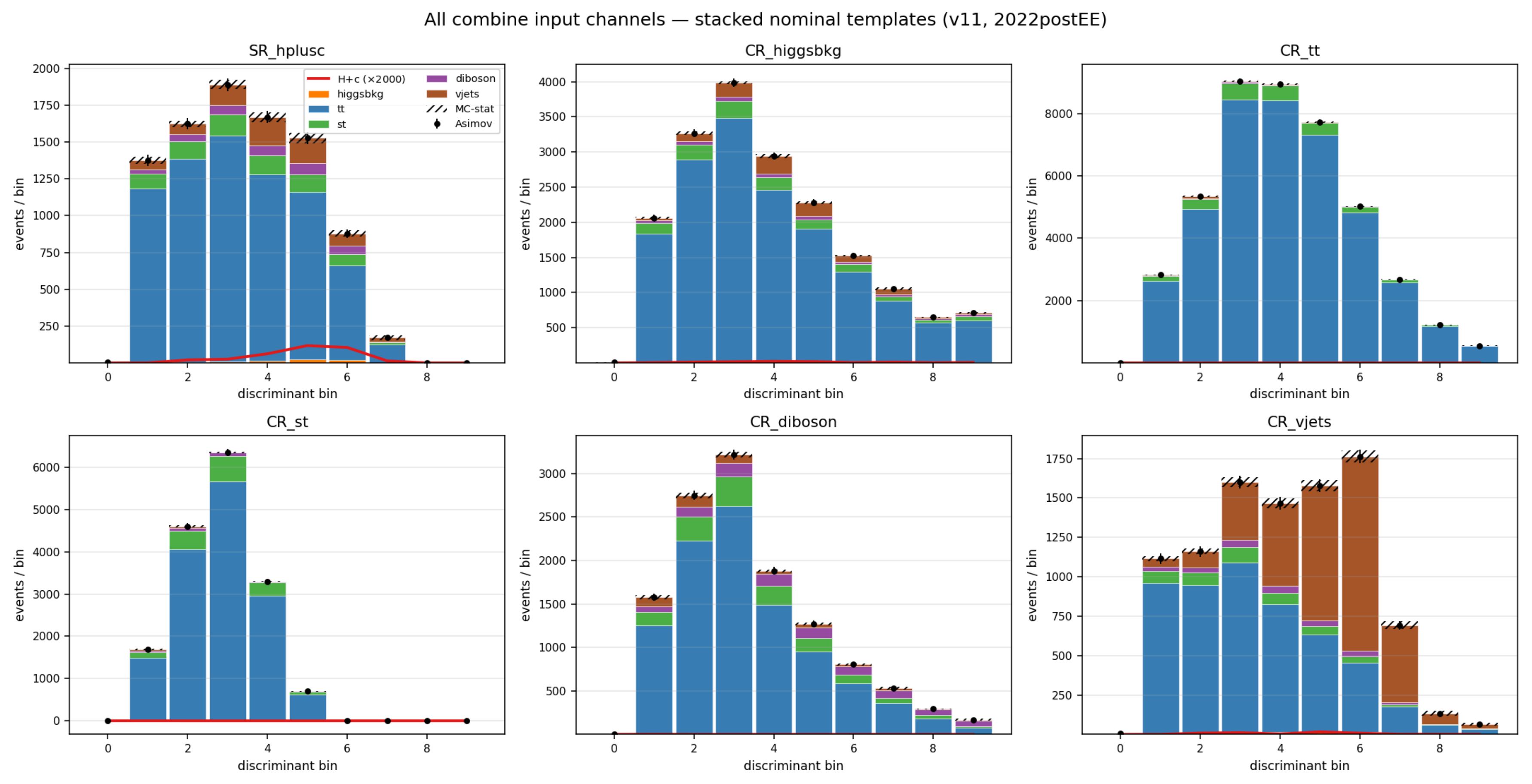
The SR is itself **82% tt**: in an eμ final state $t\bar{t}$ is dominant everywhere, so a tt-rich CR is expected, not a failure.

> **CR_tt: 94.1% purity over 44k events.** It pins the free-floating `rate_tt`, which covers 82% of the SR — the single most valuable constraint in the fit.

CR_vjets holds 59.6% of all V+jets in the fit. Without it there is no V+jets constraint from anywhere.

> Collapsing the CRs to single bins was **measured**: +33 units for three CRs, ~+50 for all five. The shapes carry real constraint, so all six channels keep 10 bins.

Templates entering the fit



6 channels × 6 processes × 10 bins. **CR_vjets** (bottom right) is the only region where V+jets (orange) is visible — it carries 59.6% of all V+jets in the fit.

3 · c-tagging

From working points to a plane

PNet gives **two** discriminants per jet: **CvL** (charm vs light) and **CvB** (charm vs bottom). A single working-point cut throws most of that away.

The calibration instead partitions the **whole plane** into **11 categories** `L0, C0–C4, B0–B4` — including the untagged `L0` bin.

property	value
axes	CvL, CvB — **verified sufficient**, no third variable needed
categories	11: `L0`, `C0` – `C4`, `B0` – `B4`
index	computed from CvL/CvB **already stored per jet**
applied	natively in the processor (`CTag2DCorrector`)

> Because the scheme spans the **full plane including `L0`**, the SF machinery works even for a selection with no working point at all.

Only 7 of 11 categories are populated

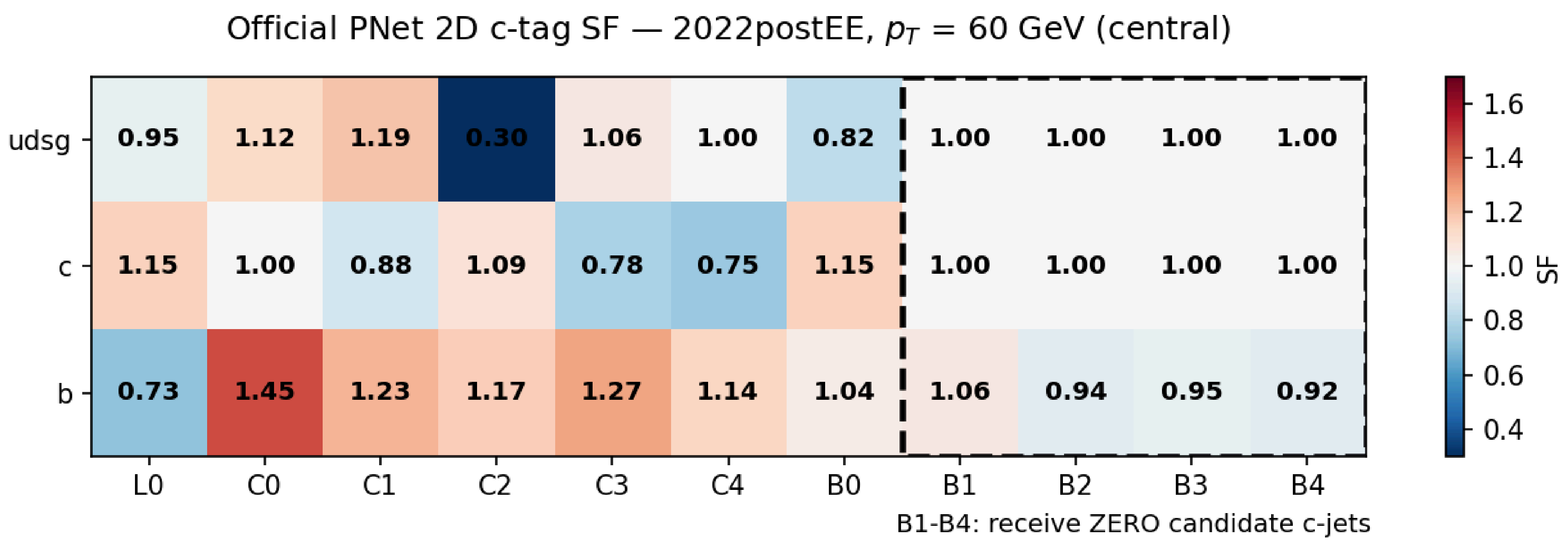
Occupancy of the five largest (`C4` and `B0` are populated but sparse):

category	jets	c (%)	b (%)	light (%)
`L0`	130,694	13.5	4.7	**81.8**
`C0`	188,702	28.2	6.2	65.7
`C1`	170,043	58.0	7.9	34.1
`C2`	17,823	58.6	33.4	8.0
`C3`	1,082	30.6	**57.9**	11.5

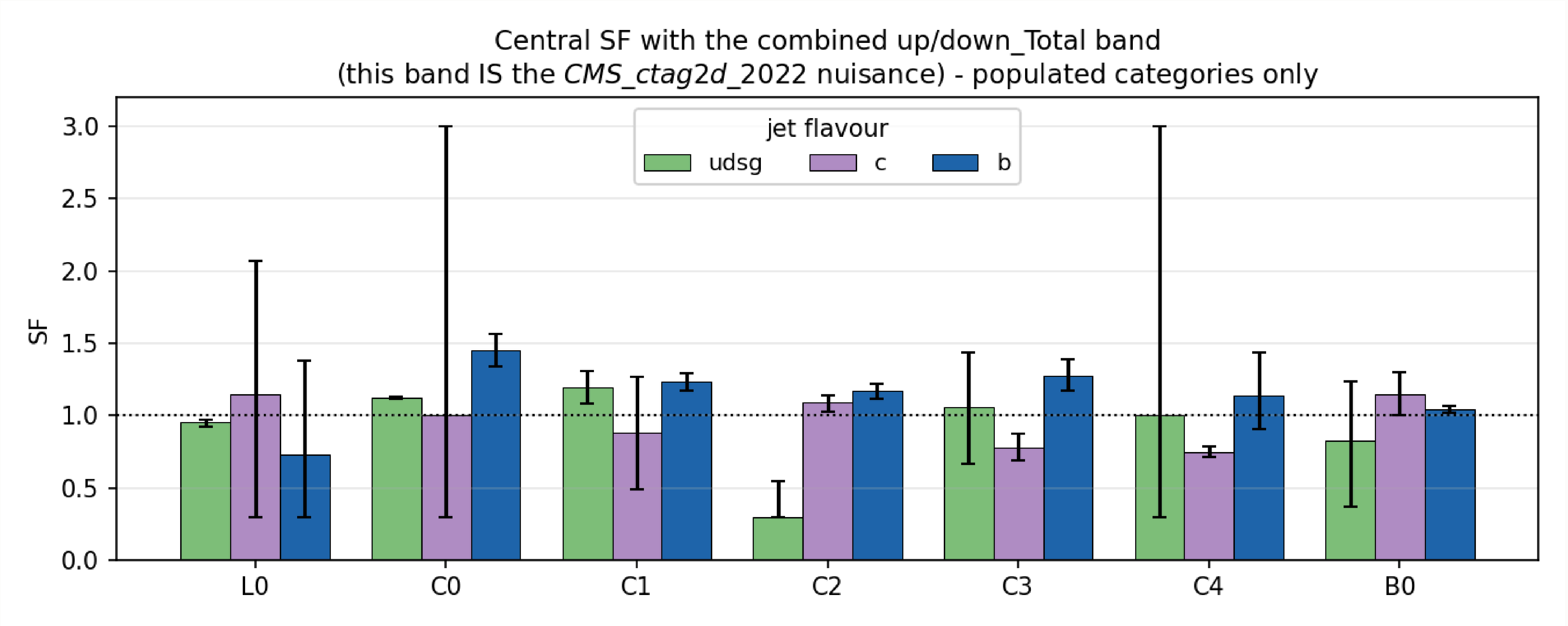
The scheme targets a broader phase space than ours — the b-rich `B1`–`B4` categories are empty after the eμ + c-jet selection.

> Category index is computed from CvL/CvB already stored per jet — **no NanoAOD reprocessing was ever needed**.

The scale factor matrix



The uncertainty band is the nuisance



What the calibration costs

configuration	limit
no SF	1150
+ `CMS_ctag2d_2022`	**1164**

Applying the calibration **worsens** the limit by 14 units — as it must.

A scale factor adds a nuisance. The point is correctness.

Currently **one nuisance** covering the whole plane. Decorrelation is pending.

4 · Negative-weight reweighting

full detail: `negrw-deck.pdf` *(57 slides)*

The estimator is a cancellation

aMC@NLO events carry **signed** weights. The yield estimator

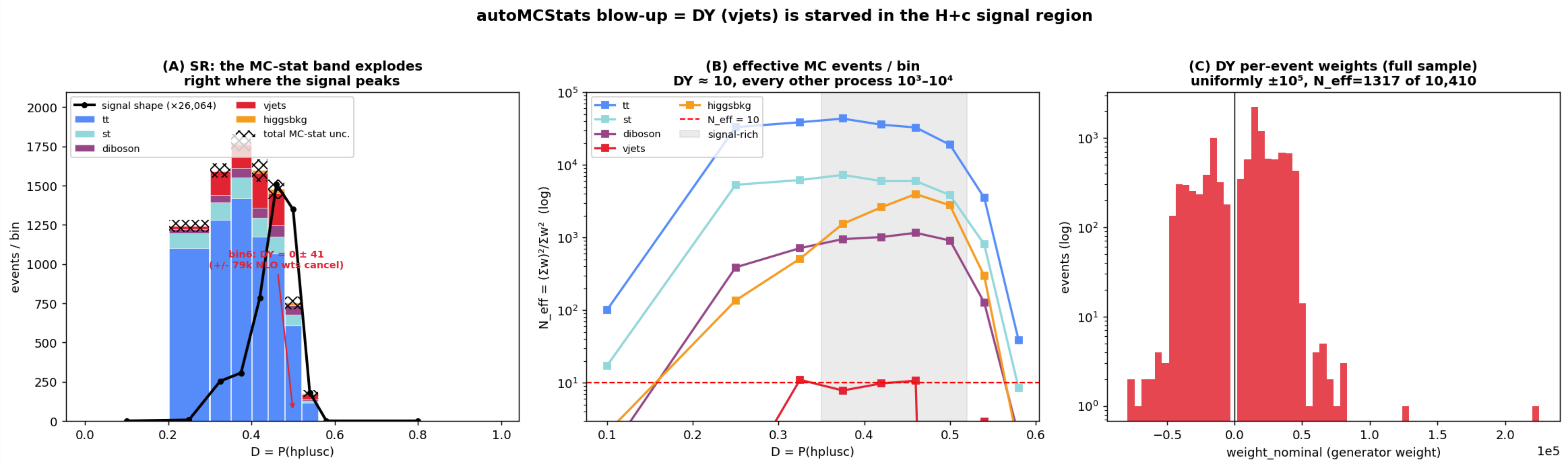
$$\hat{N}_B = \sum_{i \in B} w_i, \qquad w_i = \pm|w_i|$$

is a **difference of two large numbers**.

- the **expectation** is correct — the yield is unbiased
- the **variance** is not: it scales with $\sum w_i^2 = \sum |w_i|^2$, the *unsigned* statistics, while the mean is the much smaller signed sum
- effective statistics collapse: $N_{eff} = (\sum w)^2 / \sum w^2 \ll N$

> **16.4%** of our V+jets events carried a negative weight. In a sparse bin the positives and negatives can cancel **completely** — leaving zero content with a large finite error.

V+jets was starved exactly under the signal



(A) the MC-stat band explodes where the signal peaks — one bin was DY $= 0 \pm 41$ from ±79k weights cancelling · **(B)** N_{eff} per bin: V+jets ≈10, every other process 10^3–10^4 · **(C)** DY per-event weights, uniformly $\pm 10^5$

The method — an algebraic identity

Write the signed generator density as a difference of two positive densities:

$$\mathrm{PDF}(\vec{x}) = a\,\mathrm{PDF}_+ - b\,\mathrm{PDF}_-, \qquad a - b = 1$$

With $P_+(\vec{x}) = a\mathrm{PDF}_+/(a\mathrm{PDF}_+ + b\mathrm{PDF}_-)$:

$$\mathrm{PDF}(\vec{x}) = \underbrace{(2P_+(\vec{x}) - 1)}_{g(\vec{x})} \cdot (a\mathrm{PDF}_+ + b\mathrm{PDF}_-)$$

The right factor is the **unsigned** density — what $\sum |w|$ samples.

> So $\sum |w|\, g$ and $\sum w$ estimate **the same distribution**. Yield-preserving by construction, not an approximation.

Exact for the *true* P_+. We use $\hat{P}_+$ from a finite classifier — so closure becomes an **empirical** property that must be measured.

Training region — train loose, infer tight

$g(\vec{x})$ is a **generator property**; it does not know about the analysis selection.
So we train on a far larger sample than the SR.

	region
train	all base cuts + veto of the eμ SR topology
infer	the tight eμ signal region

> **Disjoint by construction.** An event that is not "exactly one eμ pair" can never be an SR event — no event-id bookkeeping needed.

Rejected alternatives: inverting MET (kinematically softer → extrapolation) and inverting the c-jet requirement (flips into the *opposite* gen corner).

Inputs — 20 generator-level features

group	variables
LHE event	`lhe_njets`, `lhe_nb`, `lhe_nc`, `lhe_nuds`, `lhe_nglu`, `lhe_npnlo`
LHE kinematics	`lhe_ht`, `lhe_htincoming`, `lhe_vpt`, `lhe_alphas`
gen partons	multiplicity, `n_pt20/100/200`, incoming PDG IDs
leading partons	`genparton1/2_pt`, `genparton1/2_eta`

> **All generator-level.** The paper explicitly **rejects reco-level variables** — closure is only guaranteed on quantities the generator itself sampled.

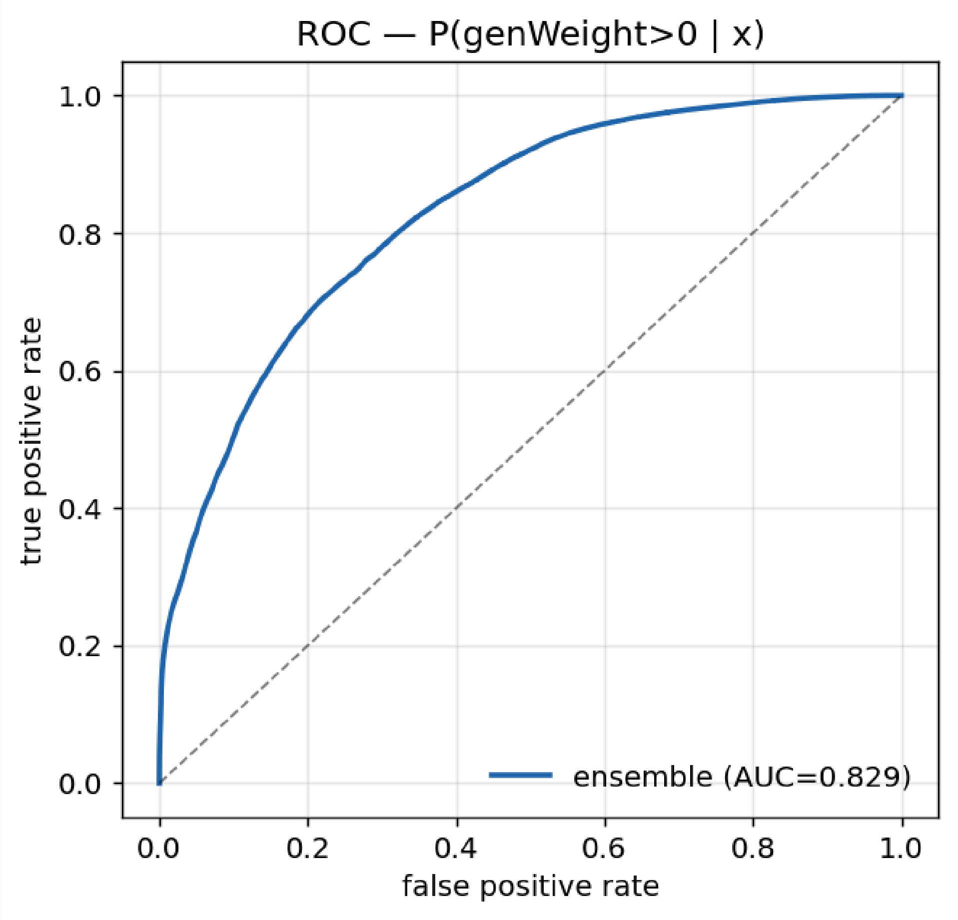
The two weight classes separate visibly in nearly every one of these features.

The classifier

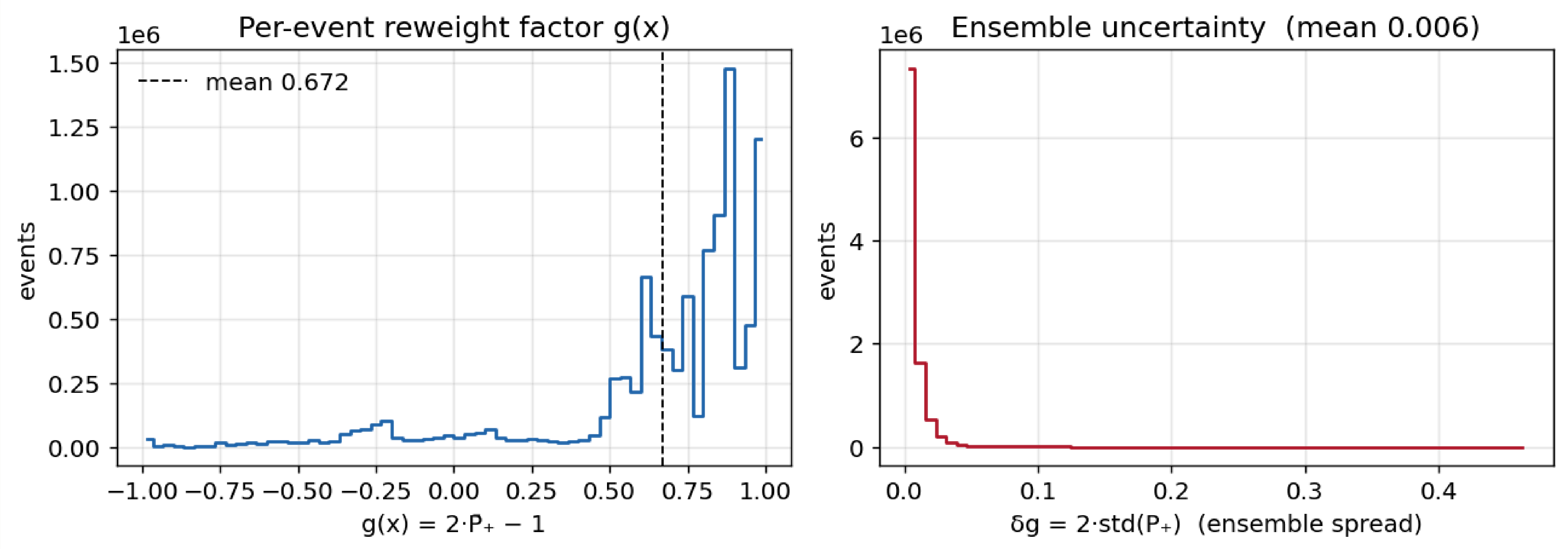
setting	value
model	`HistGradientBoostingClassifier` (scikit-learn)
inputs	20 generator-level (`lhe_*` , `genparton_*`)
training events	9.8M
ensemble AUC	**0.829**

> AUC 0.83 on an **intrinsically stochastic** target — the weight sign is not a deterministic function of the kinematics, so a perfect classifier cannot exist even in principle. 0.83 means the generator-level features genuinely carry the information the NLO subtraction encodes.

Classifier ROC

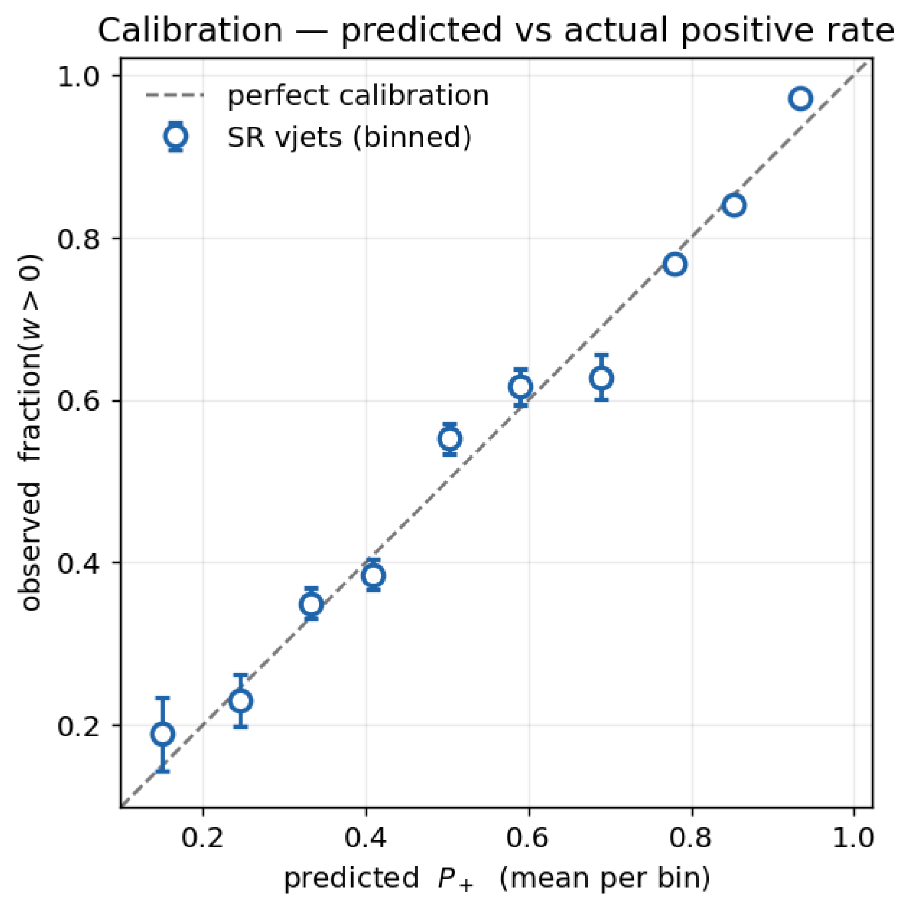


The reweight factor and its spread



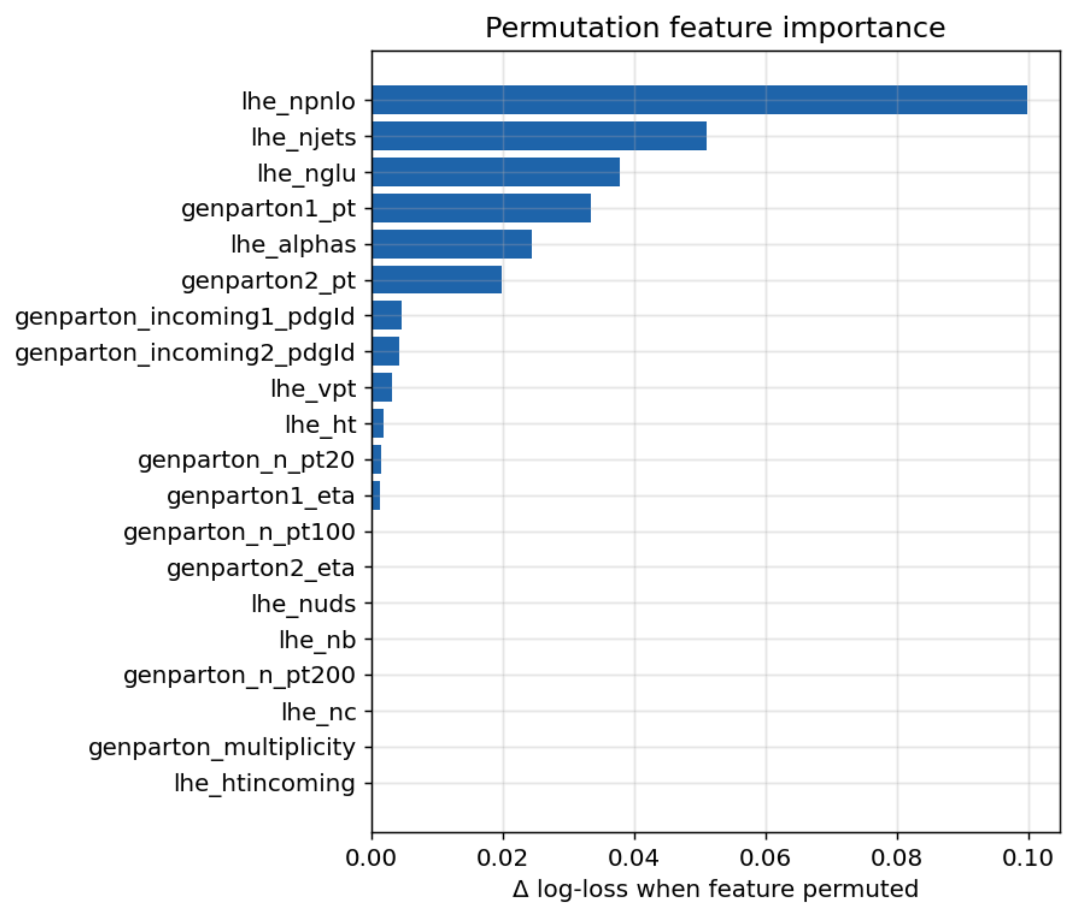
quantity	value	reading
g mean	**0.672**	$= 2 \times 0.836 - 1$, matching the global positive fraction
g range	$[-0.991,\ 0.993]$	inside the physical $[-1, 1]$ — no pathological events
δg mean / max	**0.006** / 0.467	20-model ensemble agreement is tight

Is P_+ calibrated outside the training region?



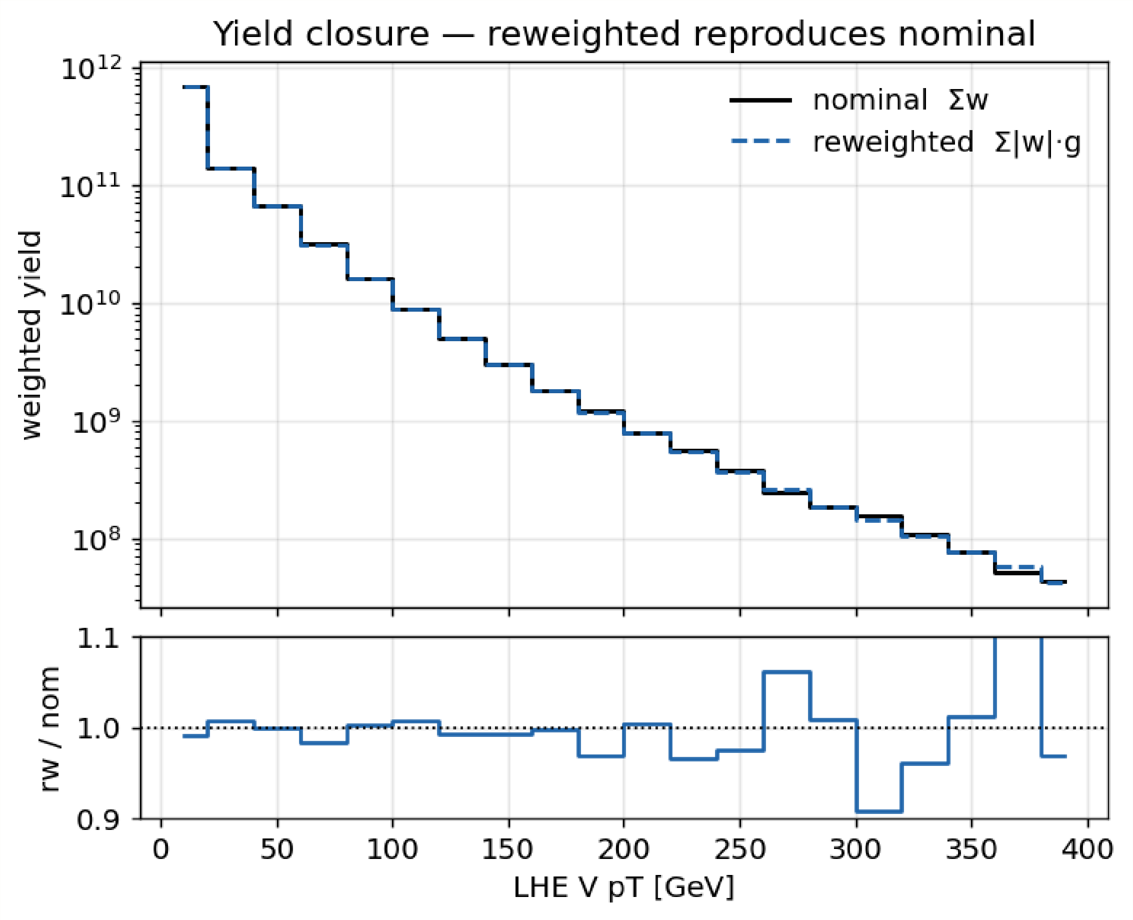
Bin SR events by **predicted** P_+, plot the **observed** fraction with $w > 0$.
Points land on the diagonal across the full range ($P_+ \approx 0.15 \to 0.93$) —
the classifier is calibrated where it is applied, not only where it was trained.

Which features carry the information



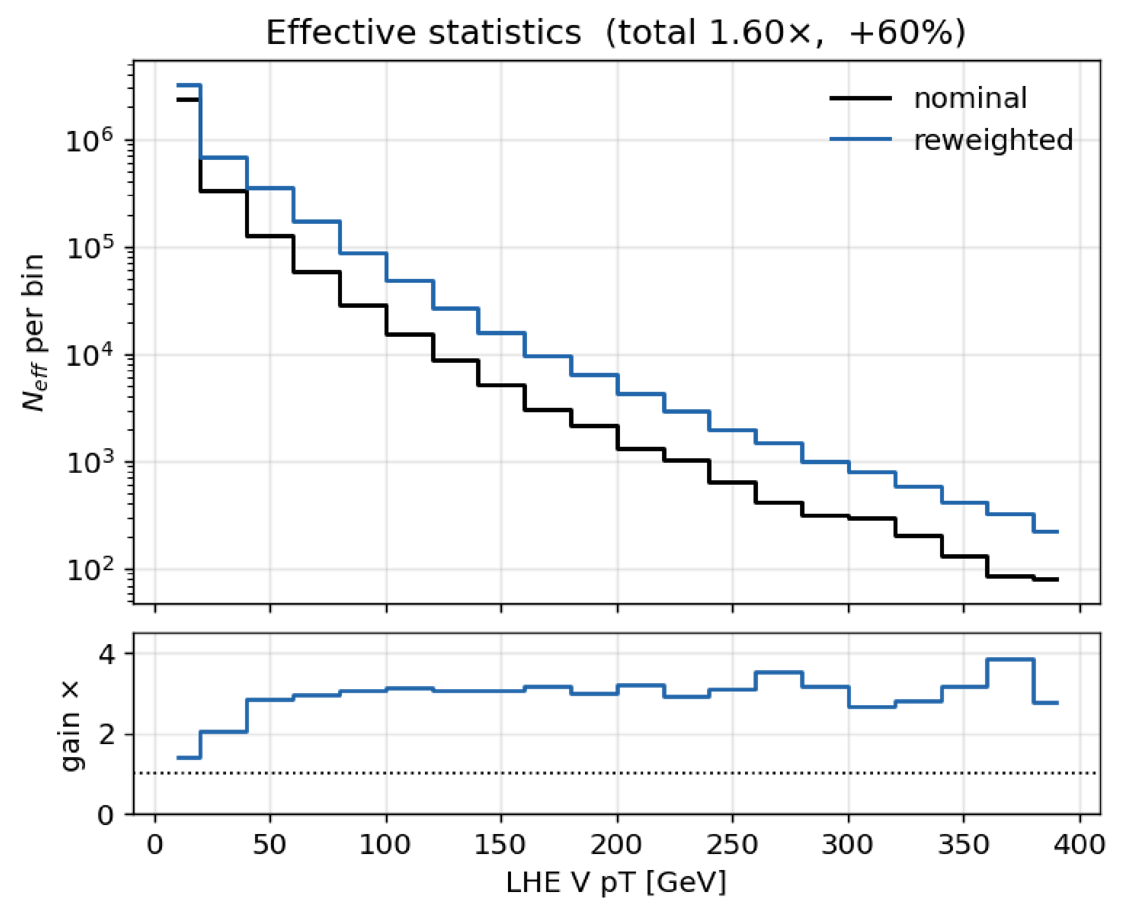
Permutation importance: the increase in log-loss when a feature is scrambled.

Closure



Training-region closure **0.994** on 9.8M events — the reweighting removes variance and nothing else.

The gain



Integrated gain **1.6×**; per-bin ratio **≈3×** in the signal-rich bins.

The honest caveat

The SR closure carries an offset. The signal region is a small, kinematically-biased corner of the space the classifier was trained on, so a finite $\hat{P}_+$ does not close perfectly there.

Why this is acceptable, stated plainly:

- reweighting fixes **variance, not yield** — exact closure is what the identity guarantees for a *perfect* classifier, not a finite one
- it **keeps the entire benefit**: N_{eff} comes from reduced per-bin variance, which survives the offset
- the residual is covered by `CMS_negrw_vjets`, which profiles to **0.0%**

> *This caveat is our extension, not the paper's* — their closure was clean because their evaluation region was not this starved.

5 · W+jets jet-binned samples

Inclusive → jet-binned

AN-23-102 §2.3 rejects the inclusive NLO sample outright:
"not used… 5 times smaller than LO and with large fraction of negative weights."

sample	xsec (pb)	events	files	neg-w
`WtoLNu-2Jets_0J`	55,760	678M	3,432	10.2%
`WtoLNu-2Jets_1J`	9,529	523M	2,669	25.7%
`WtoLNu-2Jets_2J`	3,532	345M	2,135	34.7%
sum	**68,821**	**1,546M**	**8,236**	
old inclusive	*67,710*	*282M*	*381*	*16.1%*

Cross sections from **XSDB**; sum within **+1.6%** of the inclusive — normal NLO merging spread, not a double-count. Measured negative-weight fractions match XSDB to ~1%.

> XSDB labels these `accuracy: "LO"` — **wrong**, auto-populated. `amcatnloFXFX` is NLO, proven by the 10–35% negative weights, impossible at LO.

Why the gain is not simply 5.5×

Raw events rise **5.5×**, but the useful quantity is effective statistics:

sample	n_{eff}/N	equivalent lumi
0J	0.635	7.7 /fb
1J	0.237	13.0 /fb
2J	0.094	9.1 /fb
combined		**29.8 /fb**
inclusive	*0.460*	***1.9 /fb***

The inclusive sample spends most of its cross section on **0-jet** events, which almost never survive the ≥1 c-jet requirement.

> Of the 4,851 surviving SR events, **2J alone contributes 69%** — despite being the smallest sample. The jet-binned samples are enriched in exactly what the selection needs.

W+jets result: 1160 → 1034

	before	after
expected UL	1160	**1034**
V+jets n_{eff} (SR)	280	**1170**
V+jets stat. error	5.98%	**2.92%**
V+jets rate	1508	1523

The rate barely moved while the statistical error halved — the signature of a statistics gain, not a normalisation change.

Was the gain really W+jets?

`tt` (+4.8%) and `st` (+3.8%) also moved between the two cards — both were re-merged from scratch. Checked per-process:

process	n_{eff} ratio
vjets (SR)	**4.18×**
vjets (CR_st)	**9.89×**
vjets (CR_tt)	**5.86×**
tt, st, diboson, higgsbkg	0.87 – 1.13×

> **The n_eff gain is confined to V+jets.** Other processes move by ≤13% in *both* directions — re-merge jitter, not a systematic shift. Both cards were also re-verified independently: **1160** and **1034**.

6 · Systematics

The full inventory

22 shape · 9 lnN · `rate_tt` free rateParam · autoMCStats (threshold 10)

shape (weight)	shape (object shift)
`pileup`	`CMS_scale_j_2022` (JES)
`ps_isr`, `ps_fsr`	`CMS_res_j_2022` (JER)
`scalevar_muR`, `_muF`, `_muR_muF`	`CMS_scale_e_2022`, `CMS_res_e_2022`
`lhe_pdf`, `lhe_alphaS`	`CMS_scale_m_2022`, `CMS_res_m_2022`
`muon_id`, `muon_iso`	
`electron_id`, `electron_reco` ×3	**lnN**
`CMS_ctag2d_2022`	`lumi_13p6TeV`, `xsec_st/diboson/vjets/higgsbkg`
`CMS_negrw_vjets`	`BR_HtoWW`, `BR_Htautau`, `xsec_hplusc_4FS_5FS`
`flavor_composition_ggH` (placeholder)	

Object shifts need full reprocessing

JES/JER and lepton scale/resolution **change the MVA score**, so they cannot be applied as an event weight.

Each is a **separate parquet tree**, re-run through the full selection *and* re-scored by the network — 12 shifted directories.

> Inference coverage over every shift directory is **verified after each rebuild** — a partially-scored set would leave object-shift templates frozen at nominal.

Top-p_T reweighting

Implemented following `hh2bbww` (the framework behind AN-24-091), using the **theory-based (NNLO/NLO)** parameterisation with the Run 3 rescaling:

property	value
form	theory-based NNLO/NLO
Run 3 factor	× (0.991 + 7.5e-5·p_T)
nominal	reweighted
p_T cap	none — well-behaved at high p_T

```
sf_run2 = 0.103*exp(-0.0118*pT) - 0.000134*pT + 0.973
sf      = (0.991 + 0.000075*pT) * sf_run2
weight  = sqrt(prod(sf))       # over the two gen tops
down    = 1.0                  # "no correction" is the variation
```

The variation is **down = 1.0** ("no correction"), symmetric about the nominal.

Higgs heavy-flavour

ggH is only **13.1%** of the merged `higgsbkg` group (VBF 29.0%, ggZH 23.3%, ZH 21.0%, WH 9.1%), so a flat lnN on the group either over-penalises the other 87% or must be diluted to an average — and cannot produce a shape effect.

Implemented as a per-event weight instead, keyed on gen-jet flavour.

> **Why our Bkg-Higgs impact stays small even once activated.** Freezing it moves the limit by <1 unit, against **7.6%** in AN-23-102. Two causes: the per-event weight is **not yet active**, *and* `higgsbkg` is only **0.6% of our SR** — ours is tt-dominated, theirs is Higgs-background-dominated. It will rise, but **not to 7.6%**.

Replaced with a per-event weight, ported from HiggsDNA `Higgs_plus_HF_syst`:

```
num_HF_jets = ak.sum(genJets.hadronFlavour == 4, axis=-1)   # pT>25, |eta|<2.5
up   = where(num_HF_jets > 0, 1.5, 1.0)      # AN-23-102: 50% on ggH
down = where(num_HF_jets > 0, 0.5, 1.0)
```

> Keys on **gen-jet flavour**, so process grouping stops mattering — and it produces

MET unclustered energy

> **Implemented, not yet in the card** — it needs the shifted trees from a reprocessing pass, so it does not appear in the 22 shape nuisances above.

For Run 3 PuppiMET the shift is taken directly from the NanoAOD branches:

```
events.PuppiMET.ptUnclusteredUp / ptUnclusteredDown
events.PuppiMET.phiUnclusteredUp / phiUnclusteredDown
```

Matches HiggsDNA `MET_syst_Unclustered` — same branches, same up/down ordering.

What is implemented

systematic	implementation
`CMS_negrw_vjets`	ensemble spread of the negative-weight reweighting
`CMS_ctag2d_2022`	2D CvL/CvB SF, applied natively in the processor
`electron_reco` ×3	p_T-binned (<20, 20–75, >75 GeV)
`lhe_pdf`, `lhe_alphaS`	NNPDF replicas, MC2Hessian
`top_pt`	hh2bbww theory form — *pending activation*
`higgs_plus_c`	per-event HF weight — *pending activation*
MET unclustered	object shift — *pending activation*

The last three are implemented and verified but **not in the current card** — each needs its weight column or shifted tree from a reprocessing pass.

Deliberately excluded

item	decision
muon reco SF	**not applicable** — HiggsDNA exposes no reco key; hh2bbww registers `mu_id_sf`/`mu_iso_sf` and no `mu_reco_sf`. Two frameworks, same conclusion.
PU jet ID	absent from both frameworks for Run 3
UE / tune	requires **dedicated tune samples** — cannot be a weight
`hdamp`, `mtop`	sample-based tt modelling; absent from AN-23-102 Table 16 too

> `hdamp` and `mtop` are standard Run 3 tt modelling terms. Both need alternative samples. Recorded as a **documented decision**, not an oversight.

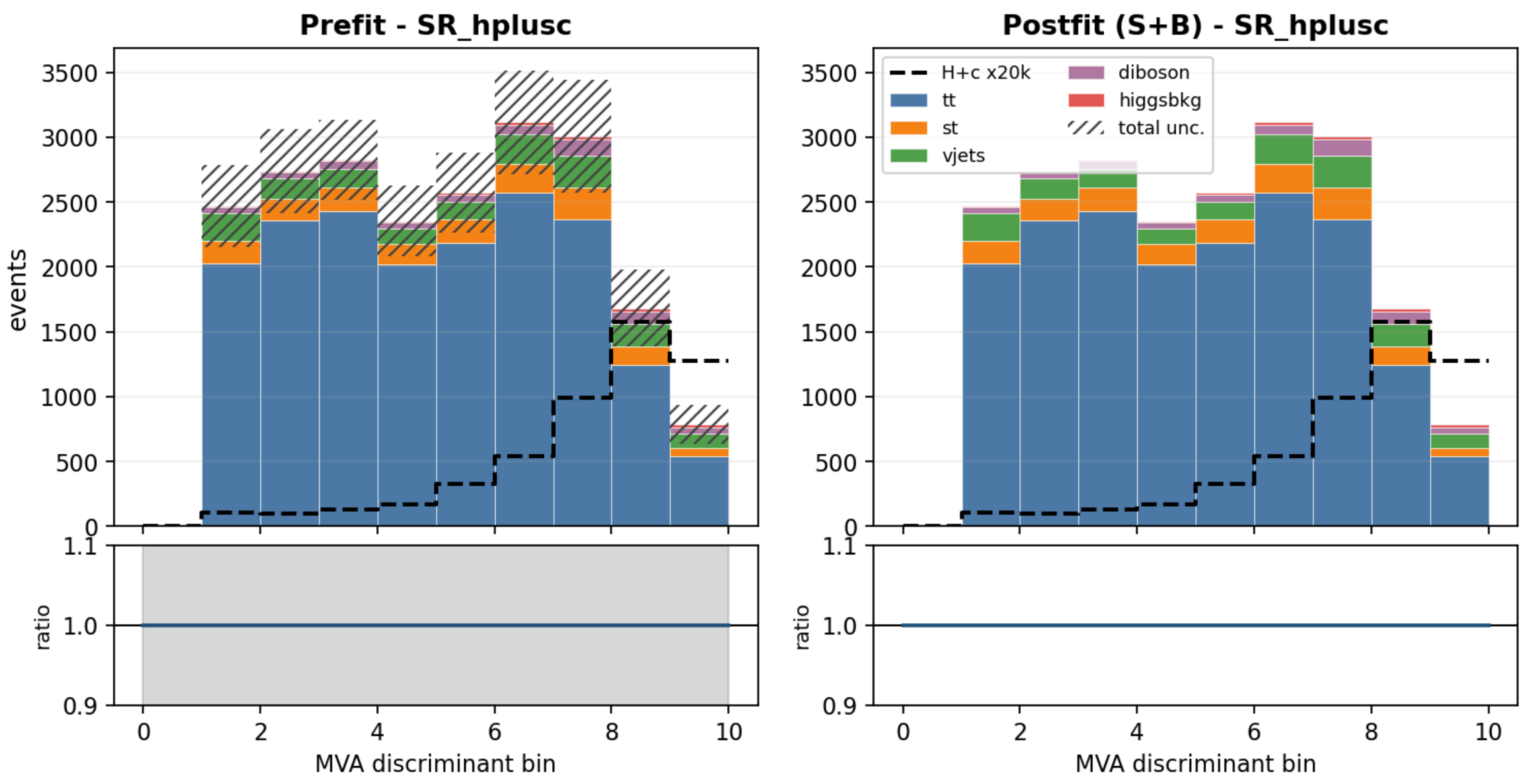
What moves the limit

Each nuisance frozen in turn; **Δ is the improvement in the expected limit.**

frozen	limit	Δ	% of 1034
all constrained (stat-only)	**641**	**393**	**38.0%**
theory shapes (scale + PS + α_S)	883	151	14.6%
`xsec_hplusc_4FS_5FS`	921	113	10.9%
`scalevar_muF`	944	90	8.7%
autoMCStats (all)	956	78	7.5%
`CMS_ctag2d_2022`	964	70	6.8%
`rate_tt`	1011	23	2.2%
JES	1023	11	1.1%

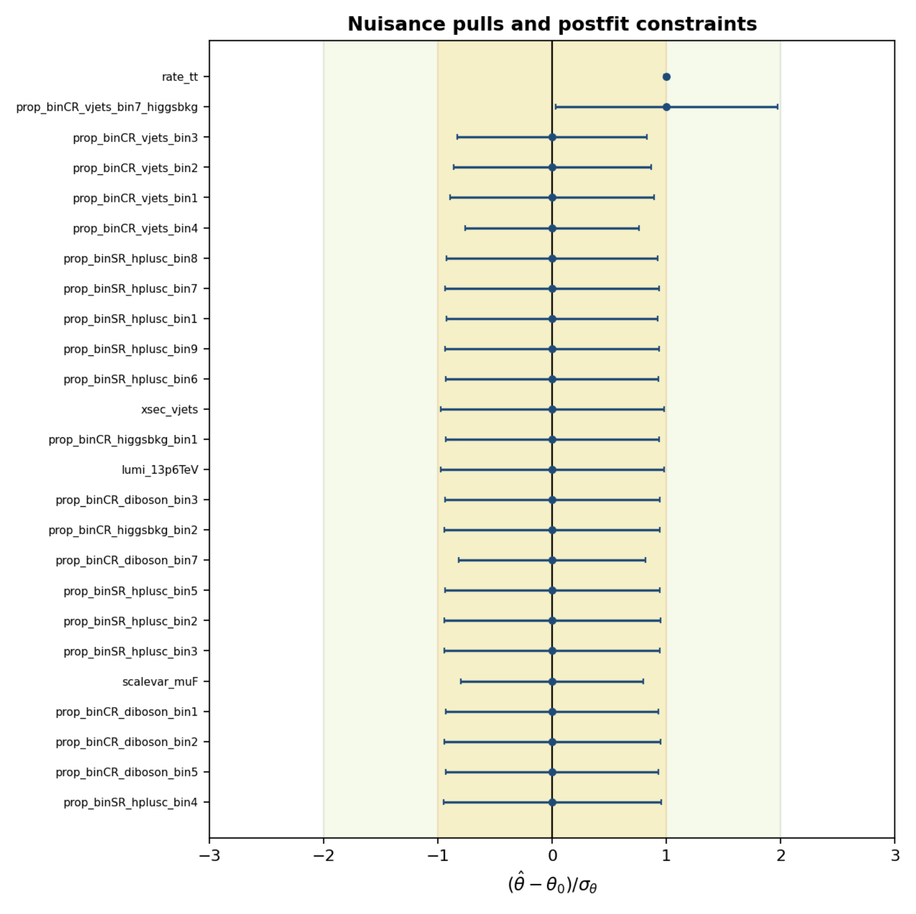
> **autoMCStats is no longer the leading term.** It was dominant before the W+jets change; at 7.5% the analysis is no longer MC-stat-limited in the way it was. Signal theory now leads.

Template composition — signal region



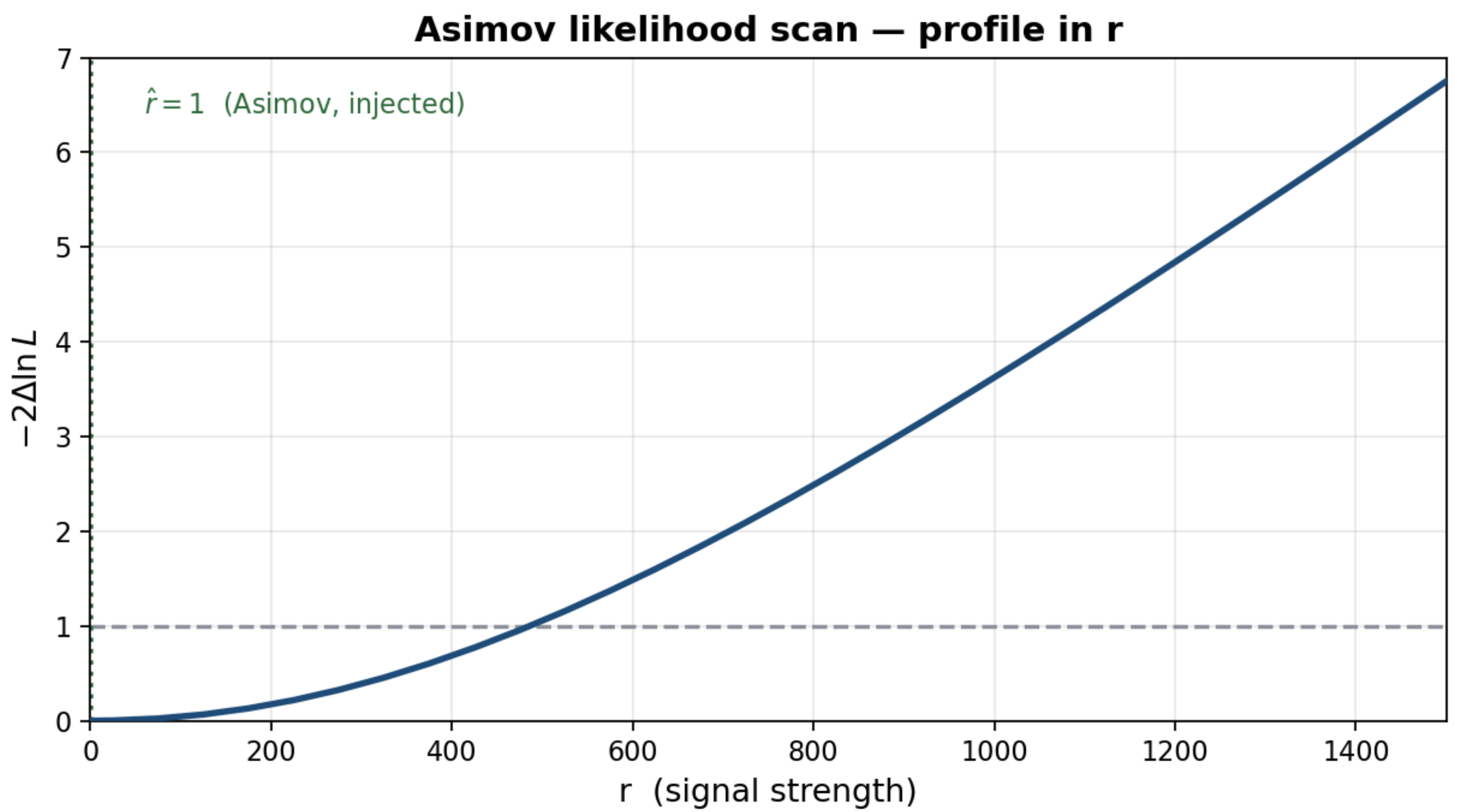
Backgrounds stacked, H+c overlaid (×20k). Hatched band = **prefit** total uncertainty, ≈14% per bin. Postfit errors are absent because combine writes a **zero postfit covariance** on an Asimov fit.

Nuisance pulls and constraints



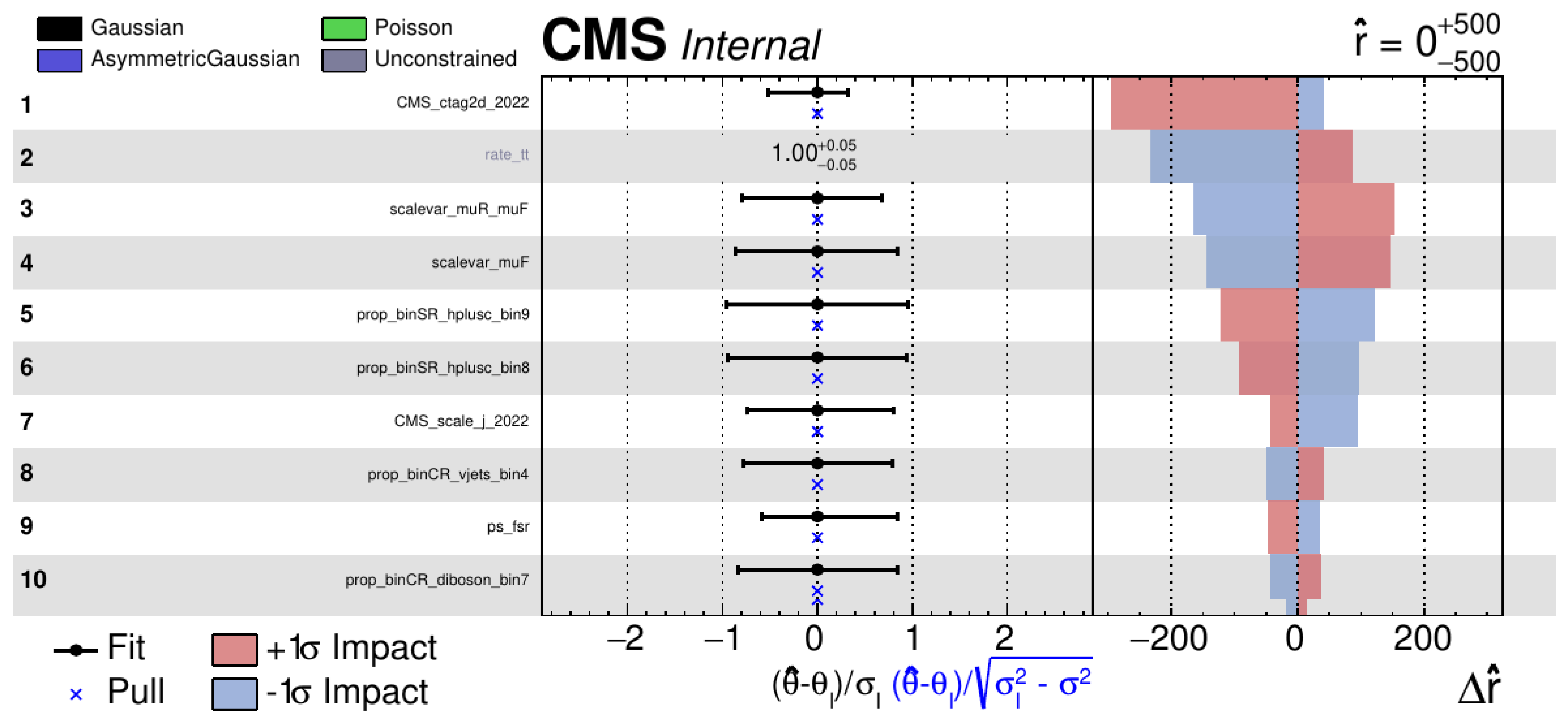
All nuisances sit at **zero pull with ~unit width** — correct for an Asimov fit, and the check that nothing is being pulled or over-constrained. **`rate_tt` is the exception**: pinned at 1.00 ± 0.05 by the 94%-pure CR_tt.

Likelihood scan



Minimum at the injected **r̂ = 1**. The dashed line is the 68% CL crossing; the quoted **1034 is a CLs upper limit**, which is a different construction and is *not* read off this curve.

Nuisance impacts — top 10



The header `r̂ = 0 +500/−500` is a **plotImpacts rounding artefact** — the actual fit returns **r̂ = 0.99996, −513/+483**, i.e. exactly the injected Asimov value.

Reading the impacts — the asymmetries

Left panel — the pull $(\hat{\theta} - \theta_0)/\sigma_\theta$. All pulls sit at zero: this is an **Asimov** fit, so by construction nothing is pulled. A non-zero pull here would mean the fit is being dragged, and would need explaining.

Right panel — $\Delta\hat{r}$ when each nuisance is moved by $\pm 1\sigma$.

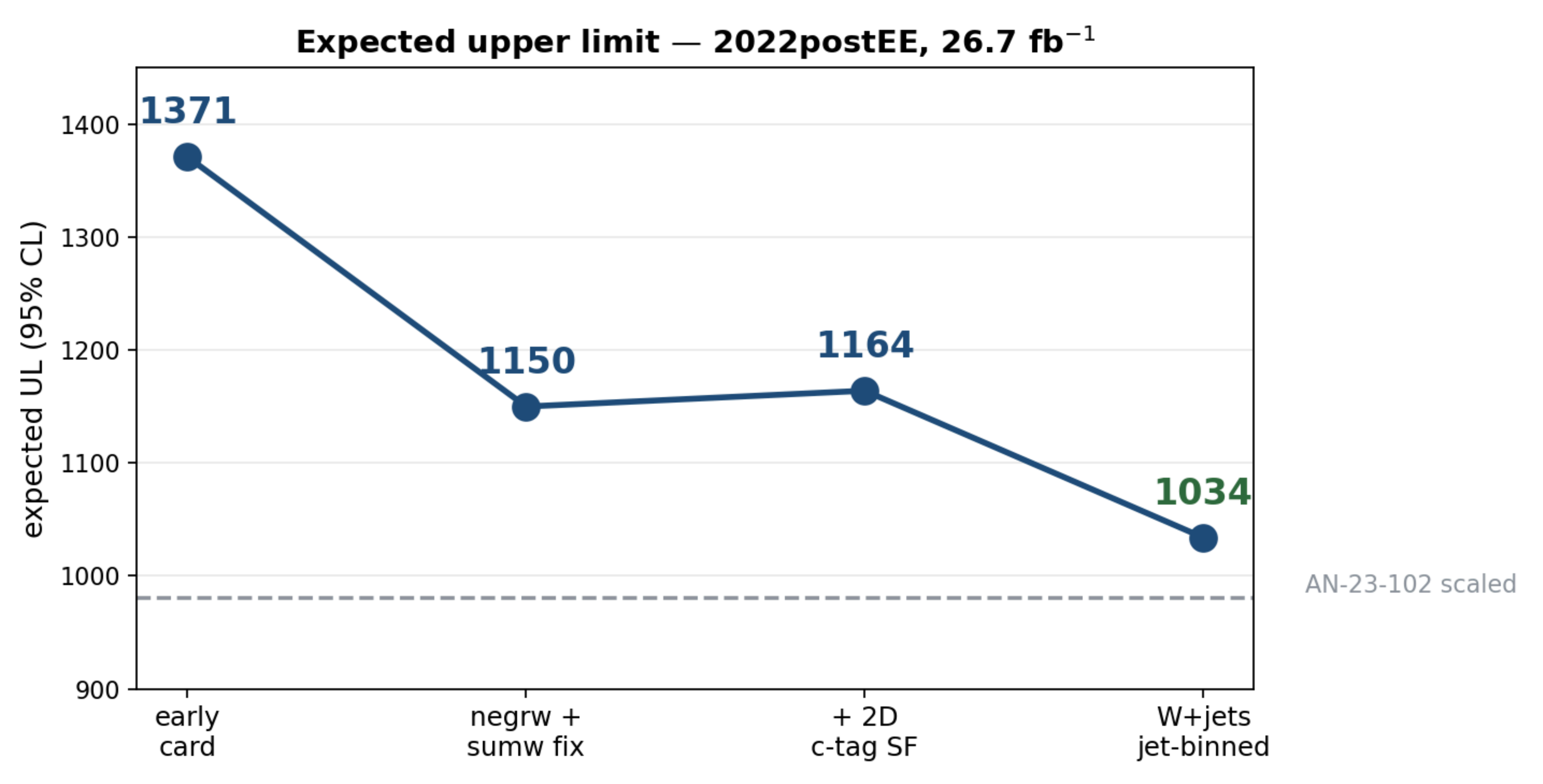
> **Red (+1σ) and blue (−1σ) are not mirror images.** Three sources of asymmetry:

source	why
asymmetric lnN	`xsec_st` is `0.9873/1.0167` — the up and down variations differ by construction
template shape	up/down shifts of JES or ctag move events between bins non-linearly
re-profiling	pushing one nuisance lets the others re-fit, and they respond differently in each direction

> The **fit range matters**. An earlier run with `--rMin 0` clipped every −1σ impact at the boundary and produced a one-sided plot. A symmetric range around $\hat{r}$ is required for the impacts to be meaningful.

7 · Status and outlook

The road so far



Note the suppressed zero on the y-axis — the c-tag step is a genuine but small **+14** increase, not the spike it appears to be.

Where we stand

	limit
AN-23-102 scaled to 26.7 fb^{-1}	**1144**
this analysis	**1034**
our statistical only	**641**

Better than the published analysis scaled to our luminosity — 1034 vs 1144, a **9.6% improvement**, on **5× less data**.

The scaling is indicative, not rigorous. $503 \times \sqrt{138/26.7} = 1144$ assumes a purely statistics-limited extrapolation. AN-23-102's Table 17 puts its statistical term at 73.8%, so it is *not* fully statistics-limited — the true scaled value would be somewhat **worse** than 1144, making our margin **larger** than 9.6%, not smaller.

The systematic breakdown of the 1034 is **under investigation** — the full term-by-term comparison against AN-23-102 Table 17 is not yet settled.

What is still missing

No data/MC agreement plots yet — the analysis is blind and the comparison plots are still being produced.

item	status
4FS/5FS signal theory	placeholder 1.30 — needs re-derivation at 13.6 TeV
Trigger scale factors	implemented, **not yet enabled**
Higgs heavy-flavour (`higgs_plus_c`)	implemented, pending activation
top-p_T reweighting	implemented, pending activation
Dataset substitutions	W+jets p_T-binned (full AN stitch)
	DY → $Z \to \tau\tau$ filtered
	tt / single-top `-ext` samples
	`WZto3LNu`

Priorities

#	item	why
1	Re-derive `xsec_hplusc_4FS_5FS`	30% flat lnN on a statistics-starved signal
2	Enable trigger SFs	implemented, one flag
3	Activate `higgs_plus_c` + `top_pt`	proper HF and tt treatment
4	Add the missing datasets	more MC statistics
5	Decorrelate `CMS_ctag2d_2022`	currently one nuisance over the whole plane

Summary

Expected UL 1371 → 1034 · statistical-only **641**

V+jets n_{eff} **280 → 1170**

Within ~5% of the Run 2 analysis scaled to our luminosity

Remaining: 4FS/5FS re-derivation, trigger SFs, dataset substitutions